DS	**Session**	Stack	**Question Information**	Level 1 Challenge 42

Question description

Rajinikanth organized a technical round interview in an Animation company for the set of computer science candidates.

the task is to implement stack operations for two stacks and merge the stacks into one.

Get two sets of stack elements and reverse them, then merge them into one stack.

Rajinikanth has given the deadline of only 15 minutes to complete the problem.

Can you Help the candidates to complete the problem within the specified time limit?

Function Description

a) push(): Adds the new item at the beginning of the linked list using the first pointer.
b) pop(): Removes an item from the beginning using the first pointer.
c) merge(): Links the first pointer second stack as next of the last pointer of the first list.

Constraints

$0 < n, m < N$

$1 < arr[i] < 1000$

Input Format:

First-line indicates n & m, where n is the number of elements to be pushed into the stack and m is the number of pop operations that need to be performed

The next line indicates the n number stack elements

Output Format:

First-line indicates the top of the element of the stack

the second line indicates the top of the element after the pop operation

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
5 8
9 8 7 6 5
5 6 7 8 3 11 2 3
```

EXPECTED OUTPUT

```
5 6 7 8 9 3 2 11 3 8 7 6 5
```

Test Case 2

INPUT (STDIN)

```
10 4
91 18 17 16 15 15 16 17 18 13
11 12 13 14
```

EXPECTED OUTPUT

```
13 18 17 16 15 15 16 17 18 91 14 13 12 11
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
void push(int data, mystack* ms)
```

Test Case 2

KEYWORD

```
int pop(mystack* ms)
```

Test Case 3

KEYWORD

```
void merge(mystack* ms1, mystack* ms2)
```

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

```
1
```

Test Case 2

TOKEN COUNT

```
435
```

Test Case 3

NLOC

```
90
```

Code

solution.c

```c
#include <stdio.h>
#include <stdlib.h>

typedef struct Node {
    int data;
    struct Node* next;
} Node;

typedef struct mystack {
    Node* top;
} mystack;

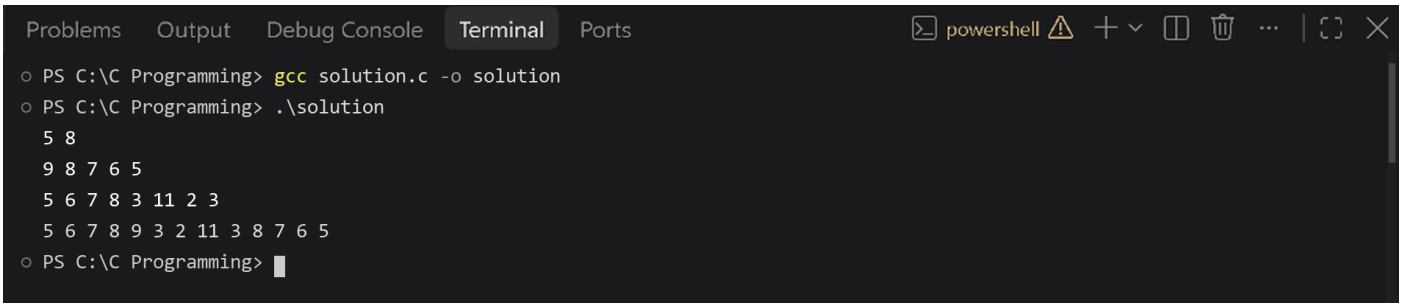
void push(int data, mystack* ms) {
    Node* temp = (Node*)malloc(sizeof(Node));
    temp->data = data;
    temp->next = ms->top;
    ms->top = temp;
}

int pop(mystack* ms) {
    if (ms->top == NULL) return -1;
    Node* temp = ms->top;
    int val = temp->data;
    ms->top = ms->top->next;
    free(temp);
    return val;
}

void merge(mystack* ms1, mystack* ms2) {
    if (!ms1->top) { ms1->top = ms2->top; return; }
    Node* cur = ms1->top;
    while (cur->next) cur = cur->next;
    cur->next = ms2->top;
}

int main() {
    int n, m;
    if (scanf("%d %d", &n, &m) != 2) return 0;
    mystack s1 = {NULL}, s2 = {NULL};
    for (int i = 0; i < n; i++) {
        int x; scanf("%d", &x);
        push(x, &s1);
    }
    for (int i = 0; i < m; i++) {
        int x; scanf("%d", &x);
        push(x, &s2);
    }
    merge(&s1, &s2);
    Node* cur = s1.top;
    while (cur) {
        printf("%d%s", cur->data, cur->next ? " " : "");
        cur = cur->next;
    }
    printf("\n");
    return 0;
}
```

Output


Problems
Output
Debug Console
Terminal
Ports
powershell
PS C:\C Programming> gcc solution.c -o solution
PS C:\C Programming> .\solution
5 8
9 8 7 6 5
5 6 7 8 3 11 2 3
5 6 7 8 9 3 2 11 3 8 7 6 5
PS C:\C Programming>